

Regularized Multivariate Two-way Functional Principal Component Analysis

Mobina Pourmoshir
mobina.pourmoshir@marquette.edu

Department of Mathematical and Statistical Sciences\ Marquette University

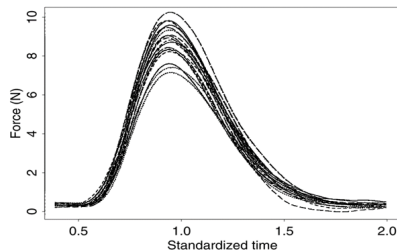


Outline

- 1 Introduction & Background
- 2 A SVD Approach for Regularized Multivariate FPCA
- 3 Two-way Regularized Multivariate FPCA
- 4 Conclusion & Future Work

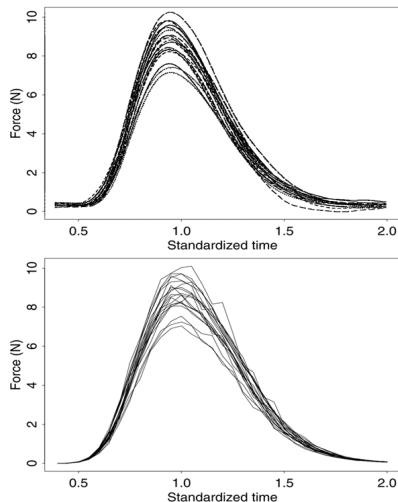
Functional Data Analysis:

- Functional data are observations that vary continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.



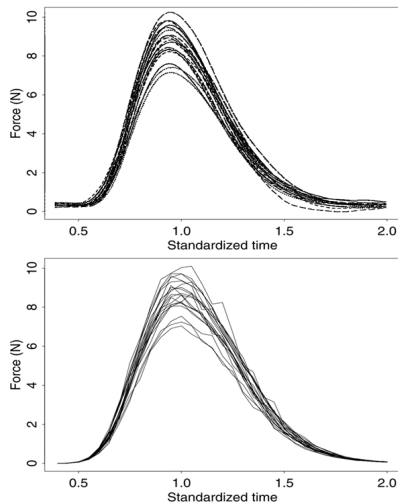
Functional Data Analysis:

- Functional data are observations that vary continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.
- These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.



Functional Data Analysis:

- Functional data are observations that vary continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.
- These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.
- Functional Data Analysis (FDA) is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.



Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation (Ramsay and Silverman 2005).

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation (Ramsay and Silverman 2005).
- Purpose: Dimension reduction and uncovering hidden patterns.

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation (Ramsay and Silverman 2005).
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation (Ramsay and Silverman 2005).
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:
 - Smoothed FPCA: Adds roughness penalties for smoothness (Silverman 1996; Huang, Shen, and Buja 2008).

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation (Ramsay and Silverman 2005).
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:
 - Smoothed FPCA: Adds roughness penalties for smoothness (Silverman 1996; Huang, Shen, and Buja 2008).
 - Sparse FPCA: Enforces sparsity for interpretability (Shen and Huang 2008; Nie and Cao 2020).

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation (Ramsay and Silverman 2005).
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:
 - Smoothed FPCA: Adds roughness penalties for smoothness (Silverman 1996; Huang, Shen, and Buja 2008).
 - Sparse FPCA: Enforces sparsity for interpretability (Shen and Huang 2008; Nie and Cao 2020).
 - Multivariate FPCA (MFPCA): Extends FPCA to multivariate functions (Silverman 1996; Happ and Greven 2018).

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation (Ramsay and Silverman 2005).
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:
 - Smoothed FPCA: Adds roughness penalties for smoothness (Silverman 1996; Huang, Shen, and Buja 2008).
 - Sparse FPCA: Enforces sparsity for interpretability (Shen and Huang 2008; Nie and Cao 2020).
 - Multivariate FPCA (MFPCA): Extends FPCA to multivariate functions (Silverman 1996; Happ and Greven 2018).
 - Regularized MFPCA: Penalties improve estimation & interpretability (Haghighi, Zhao, and Maadooliat 2023).

Functional Principal Component Analysis

- Functional PCA (FPCA): Extension of classical PCA to functional data; finds orthogonal functions capturing main variation (Ramsay and Silverman 2005).
- Purpose: Dimension reduction and uncovering hidden patterns.
- Advancements in FPCA:
 - Smoothed FPCA: Adds roughness penalties for smoothness (Silverman 1996; Huang, Shen, and Buja 2008).
 - Sparse FPCA: Enforces sparsity for interpretability (Shen and Huang 2008; Nie and Cao 2020).
 - Multivariate FPCA (MFPCA): Extends FPCA to multivariate functions (Silverman 1996; Happ and Greven 2018).
 - Regularized MFPCA: Penalties improve estimation & interpretability (Haghibin, Zhao, and Maadooliat 2023).
- Impact: More adaptable, robust, and applicable across diverse scientific and business problems.

Regularized MFPCA

Two-way Regularized MFPCA

Conclusion & Future Work

References

- Haghighin, Hossein, Yue Zhao, and Mehdi Maadooliat. 2023. "Regularized Multivariate Functional Principal Component Analysis." <https://arxiv.org/abs/2306.13980>.
- Happ, Clara, and Sonja Greven. 2018. "Multivariate Functional Principal Component Analysis for Data Observed on Different (Dimensional) Domains." *Journal of the American Statistical Association* 113 (522): 649–59. <https://doi.org/10.1080/01621459.2016.1273115>.
- Huang, Jianhua Z., Haipeng Shen, and Andreas Buja. 2008. "Functional principal components analysis via penalized rank one approximation." *Electronic Journal of Statistics* 2 (none): 678–95. <https://doi.org/10.1214/08-EJS218>.
- Nie, Yunlong, and Jiguo Cao. 2020. "Sparse Functional Principal Component Analysis in a New Regression Framework." *Computational Statistics & Data Analysis* 152: 107016. <https://doi.org/https://doi.org/10.1016/j.csda.2020.107016>.
- Ramsay, J., and B. W. Silverman. 2005. *Functional Data Analysis*. Springer Series in Statistics. Springer. https://books.google.com/books?id=mU3dop5wY_4C.
- Shen, Haipeng, and Jianhua Z. Huang. 2008. "Sparse Principal Component Analysis via Regularized Low Rank Matrix Approximation." *Journal of Multivariate Analysis* 99 (6): 1015–34. <https://doi.org/https://doi.org/10.1016/j.jmva.2007.06.007>.
- Silverman, Bernard W. 1996. "Smoothed Functional Principal Components Analysis by Choice of Norm." *The Annals of Statistics* 24 (1): 1–24.